ML for Mech Eng (CoEP)

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Outline

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About Me

Yogesh Haribhau Kulkarni

Bio:

- 20+ years in CAD/Engineering software development
- Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- Feel free to follow me at:
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Introduction

How do we learn?

- What do we do when we have to prepare for an examination?
- Study. Learn. Imbibe. Take notes. Practice mock papers.
- Thus, prepare for the unseen test.

What is Learning?

"Learning is any process by which a system improves performance from experience."

- Herbert Simon, Turing Award 1975, Nobel in Economics 1978.

What is Machine Learning?

Machine learning is a type of artificial intelligence (AI) which:

- Learns function without being explicitly programmed.
- Can grow and change when exposed to new data.

Quick Definition: *Machine learning is a field of study that gives computers the ability to learn, without being explicitly programmed.*
- Arthur Samuel, 1959
So, rather than coding each step explicitly, you give computers just some examples, and it figures out the steps.
Another Definition of Machine Learning: Machine learning is manifestation of statistical learning: implemented through software.

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Quick Check: Introduction

Think About It

A spam filter gets better at catching spam the more labeled emails you show it. A phone's autocorrect makes the same suggestion for the same typo every single time, no matter how often you type it. Which one is actually "learning", and why?

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Quick Check: Introduction

Answer

The spam filter is learning: its parameters update from experience (labeled examples), and its performance (accuracy) measurably improves. The static autocorrect isn't learning, even though it looks smart; its rules are fixed and don't change with experience. That distinction, fixed rules vs. experience-driven improvement, is what separates traditional programming from machine learning.

Why Machine Learning Matters?

Goal of Statistical learning

Dependent variables need to predicted or estimated in terms of independent variables.

- Data in control: independent variables.
- Data not in control: dependent variables.

Example

Goal: to measure sales based on the advertising budget allocated for TV, Radio, and Print.

- Can control: budgets of TV, Radio, and Print.
- Cannot control: how they will impact the sales.
- Express dependent (sales) as function of independent (advertising budget).
- Want to uncover this hidden relationship.
- Statistical learning reveals hidden data relationships.
- Relationships between dependent and independent data.

Mathematical Definition of Machine Learning

Machine Learning comes up with a Model given inputs and targets.

- Input data is available.
- Input data is transformed to get output.
- Output: something that needs to be predicted or estimated.
- Transformation engine is called Model or function.

Model entities

For $income = c + \beta_0 \times education + \beta_1 \times experience$

- Inputs: Education and experience, also called as features or attributes or dimensions or variables.
- Mathematical entities added to input data, are Parameters.. c, β_0 and β_1 are parameters
- Income is target, also called as outcome or class.

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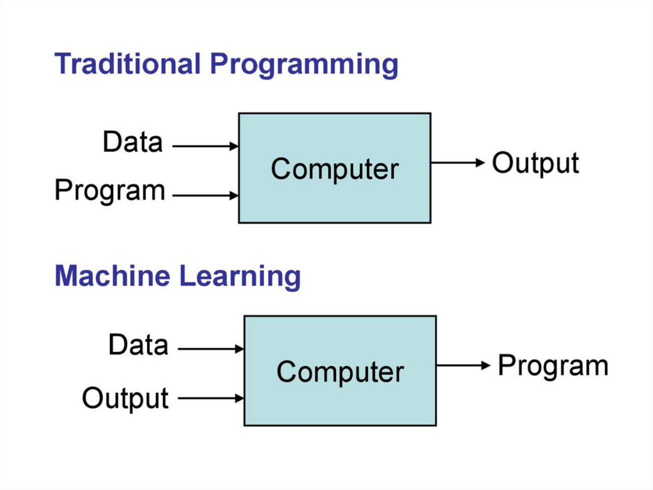
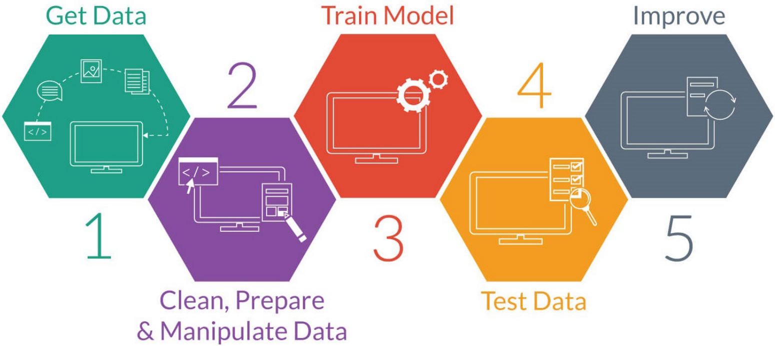
Traditional vs. Machine Learning?



Why Machine Learning?

- Problems with High Dimensionality
- Hard/Expensive to program manually
- Techniques to model 'ANY' function given 'ENOUGH' data.
- Job $$$

Machine Learning Process



(Reference: The Role of Big Data in Strengthening Machine Learning Projects - Techno FAQ)

Why now?

- Flood of data (Internet, IoT)
- Increasing computational power
- Easy/free availability of algorithms
- Increasing support from industries

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Hardware Revolution

- GPUs (NVIDIA H100, AMD MI300)
- TPUs (Tensor Processing Units)
- Edge AI chips (Apple Neural Engine, Google Tensor)
- Quantum computing for ML (early stages)

The ML Renaissance (2023-2025)

- Large Language Models (ChatGPT, GPT-4, Claude)
- Generative AI explosion
- Multi-modal models (text, image, video, audio)
- AI democratization through APIs
- MLOps maturity

Open Source ML Ecosystem 2025

Frameworks:

- PyTorch 2.x
- TensorFlow 3.x
- JAX
- scikit-learn 1.x

Tools:

- Hugging Face Transformers
- Weights & Biases
- MLflow
- DVC (Data Version Control)

Quick Check: Why ML Matters

Think About It

Statistical learning tries to find f such that $Y = f(X)$. If you already know the exact physics equation relating inputs to outputs (e.g. $F = ma$), do you still need machine learning?

Quick Check: Why ML Matters

Answer

No. If the true relationship is already known and exact, just use that equation directly. Machine learning earns its keep when the relationship is unknown, too complex to derive analytically, or has to be estimated from data because no closed-form model exists.

Big Data Context

The storm: The Big Data is coming

- In 2012, HBR put Data Scientists on the radar
- "The Sexiest Job of the 21st Century".
- Industry, trying to be data-driven, than manual.

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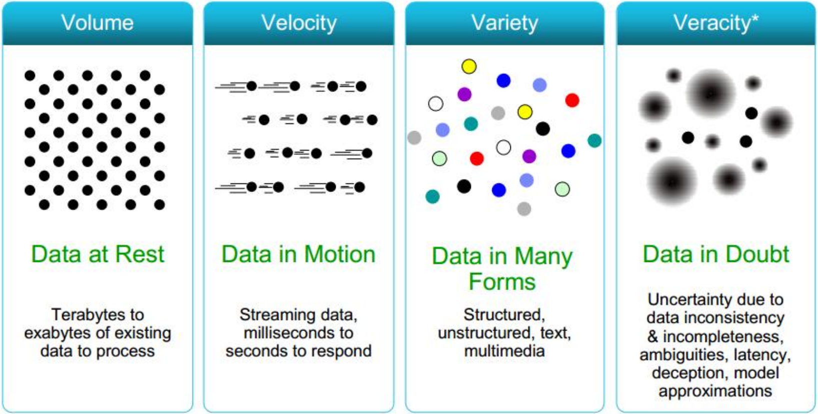
(Big) Data Characteristics



(Image Credit: http://www.rosebt.com/blog/data-veracity)

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What's the answer?

AI-ML-DL

- Machines showing intelligence of Humans
- Machine Learning: part of AI
- Logic is not programmed by hand,
- Gets emerged in training with data.

Quick Check: Big Data Context

Think About It

The "V's" of Big Data (Volume, Velocity, Variety, Veracity) are often cited as reasons ML matters now. Is having a lot of data always enough to build a good ML model?

Quick Check: Big Data Context

Answer

No. Veracity (data quality and trustworthiness) matters as much as Volume. A huge dataset that's noisy, mislabeled, or biased can make a model worse, not better ("garbage in, garbage out"). More data only helps when it's relevant and reasonably clean.

A Puzzle

How different is Machine Learning?

Maths Puzzle

Math Quiz #1 - Teacher's Answer Key

1)	2	4	5	=	3	5)	6	2	2	=	10
2)	5	2	8	=	2	6)	3	1	1	=	2
3)	2	2	1	=	3	7)	5	3	4	=	11
4)	4	2	2	=	6	8)	1	8	1	=	7

Maths Puzzle

- Letting the computer work out that relationship for you.
- ‘Learn’ to solve such problems,
- ‘Test’ with any other problem of the same type!

Quick Check: A Puzzle

Think About It

In the maths puzzle, you "learn" a pattern from a few examples and then "test" it on a new one. What could go wrong if the pattern you learned only fits the examples you were shown, but fails on new ones?

Quick Check: A Puzzle

Answer

That's overfitting: the model memorizes the specific examples instead of learning the true underlying rule, so it fails to generalize to unseen data; one of the central challenges in machine learning.

Types of Machine Learning

Three kinds of learning

- **Supervised**: learn from labeled examples (input-output pairs)
- **Unsupervised**: find hidden structure in unlabeled data
- **Reinforcement**: learn by trial-and-error to maximize cumulative reward (e.g., AlphaGo, robotics)

Supervised

- Training data with correct answers
- Both used to train the model
- Then apply unseen data on model

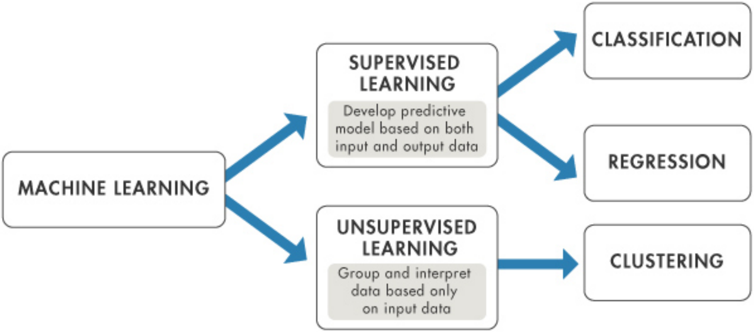
Unsupervised

- Training data with no answers
- Extract patterns, groups

Some types of algorithms

- Prediction: predicting a continuous variable from data
- Classification: assigning records to predefined groups
- Clustering: splitting records into groups based on similarity
- Association learning: seeing what often appears together

Machine Learning Algorithms



(Reference: Machine Learning in MATLAB - MATLAB & Simulink - MathWorks)

ML Algorithms: Five Key Questions

- Is this A or B? : Classification algorithms
- Is this weird? : Anomaly detection algorithms
- How much or how many? : Regression algorithms
- How is this organized? : Clustering algorithms, Dimensionality reduction
- What should I do next? : Reinforcement learning algorithms

(Ref: Brandon Rohrer's breakdown of the "5 questions data science answers")

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Classification: Overview

- **Description:** Identifying the category an object belongs to.
- **Applications:** Spam detection, Image recognition.
- **Algorithms:** SVM, nearest neighbors, random forest, Logistic Regression

Regression

- **Description:** Predicting a continuous-valued attribute associated with an object.
- **Applications:** Drug response, Stock prices.
- **Algorithms:** Linear Regression

Clustering: Overview

- **Description:** Automatic grouping of similar objects into sets.
- **Applications:** Customer segmentation, Grouping experiment outcomes
- **Algorithms:** k-Means

Dimensional Reduction

- **Description:** Reducing the number of random variables to consider.
- **Applications:** Visualization, Increased efficiency
- **Algorithms:** PCA, Singular Value Decomposition

Ranking: Overview

- **Description:** Ordering a set of items by relevance or preference, not just labeling or measuring them individually.
- **How it differs:** Classification asks "what is this?" for one item at a time; Ranking asks "which of these should come first?" for a whole set at once.
- **Applications:** Search engine results, Recommender systems, Leaderboards
- **Algorithms:** Learning-to-Rank methods (e.g., RankNet, LambdaMART), often built on Gradient Boosting

Popular Algorithms in Machine Learning

- Linear, Logistic Regression
- Decision Trees
- SVM - Support Vector Machines, Naive Bayes
- K-Means

Model Selection

- Choosing which algorithm, or family of algorithms, fits the problem – before any parameter gets fit.
- Driven by the problem type (Classification/Regression/Clustering/Ranking), data size, interpretability needs, and training-time budget.
- Rule of thumb: start simple (Linear/Logistic Regression) as a baseline, move to more flexible models (Trees, SVM, Ensembles) only if the baseline underfits.
- Not a one-time choice: gets revisited once evaluation metrics come in, and again once feature selection changes what the model has to work with.

Decision Tree: Feature Selection

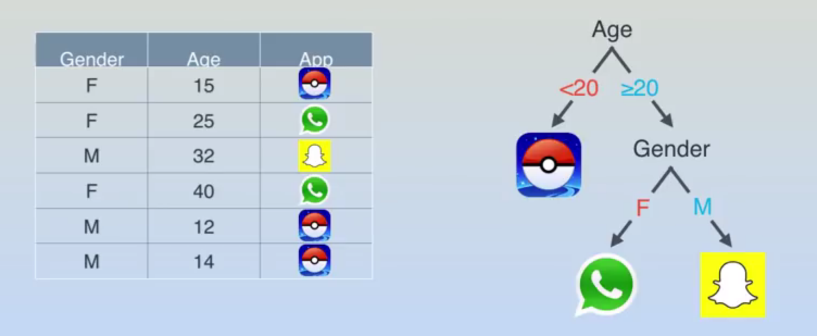
Is Gender the decider or Age?

Gender	Age	App
F	15	
F	25	
M	32	
F	40	
M	12	
M	14	

Gender is not much, but all below 20 years downloaded Pokemon Go. Age splits data best.

(Image Credit: A Gentle Introduction To Machine Learning; SciPy 2013 Presentation - Kastner, Kyle)

Decision Tree Population

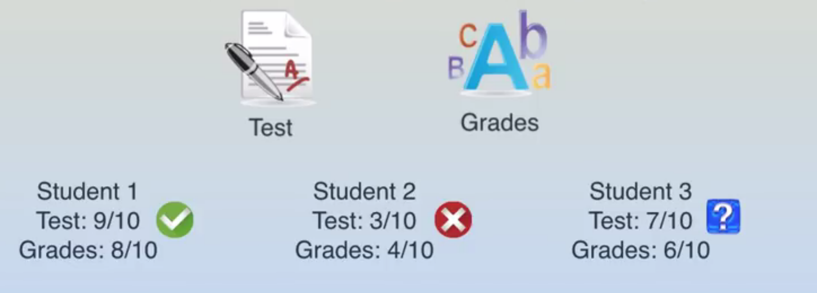


Any new person can walk through the Tree and predict.

(Image Credit: A Gentle Introduction To Machine Learning; SciPy 2013 Presentation - Kastner, Kyle)

Classification: University Admission Example

Deciding acceptance to Univ based on Test scores and grade.

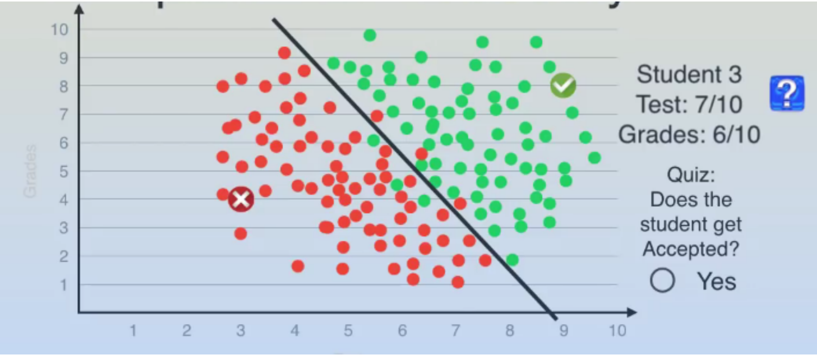


Will Student 3 get accepted?

(Image Credit: A Gentle Introduction To Machine Learning; SciPy 2013 Presentation - Kastner, Kyle)

Classification: Decision Boundary

Putting it in a grid. Test score as X and Grades as Y. Prev data shown with acceptance colored.

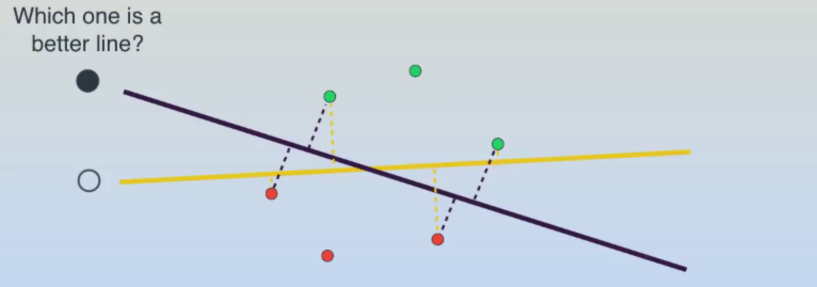


After separating line, Student 3's fate can be predicted. That's Logistic Regression.

(Image Credit: A Gentle Introduction To Machine Learning; SciPy 2013 Presentation - Kastner, Kyle)

Support Vector Machine

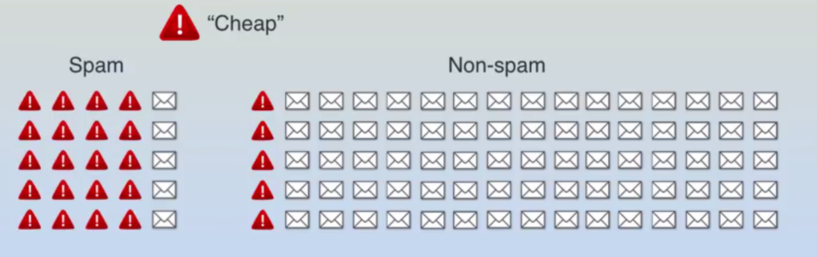
Which line separates best?



The one with max separation in the middle. That's Support Vector Machine.
(Image Credit: A Gentle Introduction To Machine Learning; SciPy 2013 Presentation - Kastner, Kyle)

Classification: Spam Detection Example

Detecting if mail is spam or not.

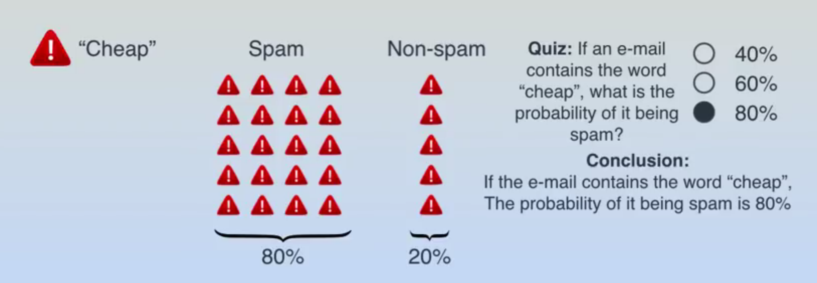


Features: mail containing "cheap".
Most past spam mails contain it.

(Image Credit: A Gentle Introduction To Machine Learning; SciPy 2013 Presentation - Kastner, Kyle)

Naive Bayes

Easy to compute probability of being a Spam, if it contains "cheap".

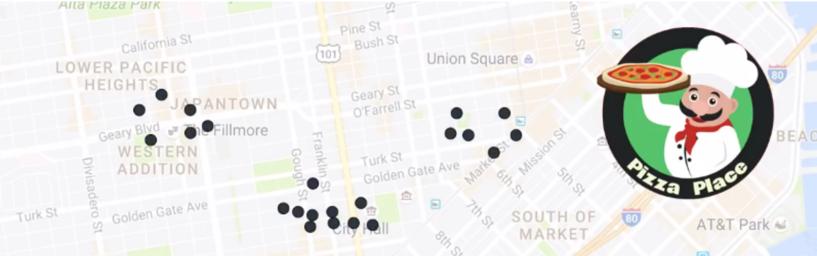


That's Naive Bayes.

(Image Credit: A Gentle Introduction To Machine Learning; SciPy 2013 Presentation - Kastner, Kyle)

Clustering: Pizza Shop Example

Wish to put 3 pizza shops in a city. Customers are plotted. What are the best locations?



Start with random locations and set ownership. Update locations. Repeat.

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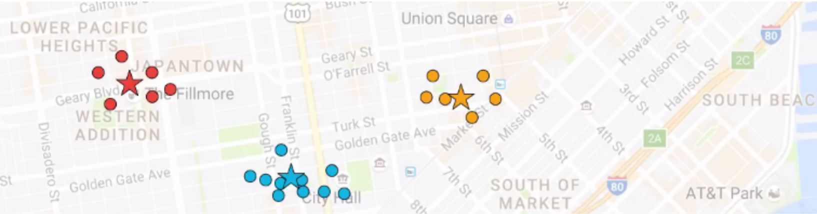
Clustering: K-Means Convergence

Locations settle at the centroids of the clusters.



That's K-means.

(Image Credit: A Gentle Introduction To Machine Learning; SciPy 2013 Presentation - Kastner, Kyle)

Quick Check: Types of Machine Learning

Think About It

You want to group customers into segments without knowing in advance how many groups exist or what defines them. Which type of learning fits, supervised or unsupervised, and why?

Quick Check: Types of Machine Learning

Answer

Unsupervised: there's no labeled "correct answer" for each customer's segment, so the algorithm (e.g. clustering) has to discover structure and groupings directly from patterns in the data.

Applications & Future

Everyday Applications of Machine Learning

- Face Recognition (Facebook)
- Spam recognition in Emails
- Recommender Systems
- Feelings Analysis, Sentiments
- Natural language: Translate a sentence from Hindi to English, question answering, etc.
- Speech: Recognise spoken words, speaking sentences naturally
- Game playing: Play games like chess
- Robotics: Walking, jumping, displaying emotions, etc.
- Driving a car, flying a plane, navigating a maze, etc.

Future Trends & Challenges

Emerging Areas:

- AGI (Artificial General Intelligence) research
- Quantum Machine Learning
- Neuromorphic computing
- AI-human collaboration tools

Challenges Ahead:

- Data scarcity for specialized domains
- Model interpretability at scale
- Energy-efficient AI
- Regulation and governance
- AI safety and alignment

Quick Check: Applications & Future

Think About It

As AI models get more powerful (bigger LLMs, more automation), do the challenges around bias, fairness, and interpretability become less important?

Quick Check: Applications & Future

Answer

The opposite: as models are deployed more widely and make more consequential decisions, the impact of bias, unfairness, or unexplainable predictions grows. Model power and responsible-AI concerns scale together, not apart.

What We Covered

- **What is ML?**: Learning patterns from data, not explicit rules
- **Types**: Supervised (labeled data), Unsupervised (hidden structure), Reinforcement (reward signals)
- **Math framework**: $Y = f(X) + \epsilon$; minimize loss to approximate f
- **Evaluation**: Cross-validation, confusion matrix, precision, recall, F1, bias-variance tradeoff
- **Toolkit**: scikit-learn's Estimator API (fit / predict / score)

Next: Data preparation and hands-on pipelines

Cool-down: Summary

SO ...

- What is Machine learning, after-all?
- Its usage in your domain?

Thanks . . .

- Office Hours: Saturdays, 3 to 5 pm (IST); Free-Open to all; email for appointment to yogeshkulkarni at yahoo dot com
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